Chapter 1

Demo problem: Flow of a fluid film down an inclined plane

The two-dimensional flow of a free surface down an inclined plane is a simple exact solution of the Navier–Stokes equations. We describe two different ways of solving the problem using either spines or a pseudo-elastic method to define the bulk mesh motion. Reassuringly, the results are the same irrespective of the method chosen.

1.1 Problem formulation

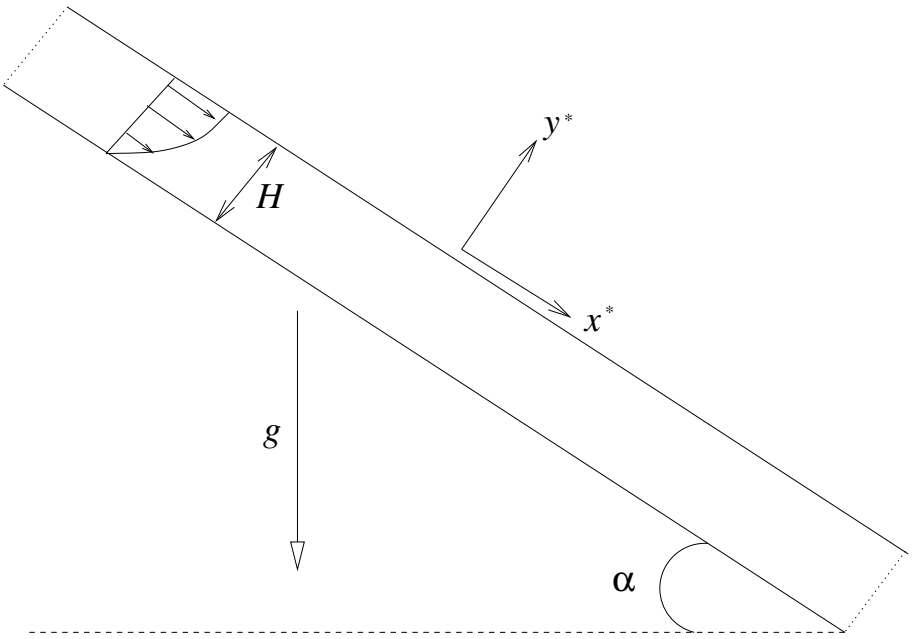


Figure 1.1 A film of incompressible viscous fluid of a given thickness flows down a plane inclined at a prescribed angle to the gravitational field.

Formulating the problem in coordinates tangential (x^*) and normal (y^*) to the plane and assuming that the flow is steady and only in the tangential direction, but independent of the tangential coordinate, reduces the momentum equations to

$$\frac{\partial p^*}{\partial x^*} = \rho\, g \sin\alpha + \mu\, \frac{\partial^2 u^*}{\partial y^{*2}},$$

$$\frac{\partial p^*}{\partial y^*} = -\rho\, g \cos\alpha,$$

where u^* is the velocity component tangential to the plane and p^* is the fluid pressure. Note that the continuity equation is automatically satisfied.
We non-dimensionalise using the only length-scale in the problem H, choosing the viscous scale for the pressure and choosing a reference velocity scale U:

$$x^* = Hx, \qquad y^* = Hy, \qquad u^* = Uu, \qquad p^* = \mu U/H,$$

and the governing equations become

$$\frac{\partial p}{\partial x} = \frac{\rho\, g H^2}{\mu U}\sin\alpha + \frac{\partial^2 u}{\partial y^2} = \frac{Re}{Fr}\sin\alpha + \frac{\partial^2 u}{\partial y^2},$$

$$\frac{\partial p}{\partial y} = -\frac{\rho\, g H^2}{\mu U}\cos\alpha = -\frac{Re}{Fr}\cos\alpha.$$

The dimensionless grouping $\rho g H^2/(\mu U)$ represents the ratio of gravitational forces to viscous forces and we choose to identify it as a Reynolds number $Re = \rho U H/\mu$ divided by a Froude number $Fr = U^2/(gH)$.
We proceed by assuming that the flow is driven entirely by the gravitational body force and that there is no additional tangential pressure gradient. Then, integrating the tangential momentum balance twice and using the boundary conditions of no-slip at the plane ($y = 0$)and that the free surface ($y = 1$) is tangentially stress-free gives

$$u = \frac{1}{2}\frac{Re}{Fr}\sin\alpha\left(2y - y^2\right).$$

Integrating the normal momentum balance and setting the reference external pressure to be zero at the free surface gives

$$p = \frac{Re}{Fr}\cos\alpha(1-y).$$

Finally, we specify a "natural" velocity scale by setting $Re/Fr = 2$, corresponding to the velocity of the free-surface for a vertical film ($\alpha = \pi/2$).
We shall assess the stability of the flat-film solution by applying a small, short-duration perturbation to the wall velocity and evolving the system in time. If the interface is stable, the perturbation should decay, if not it should grow. A linear stability analysis for this problem was performed by Benjamin (1957) and Yih (1963), who both found that for long waves in the absence of surface tension, the interface was unstable when

$$Re > \frac{5}{4}\sin\alpha.$$

(If you read the papers you will see that the Reynolds number was defined such that the average fluid downslope velocity was one; to convert to our Reynolds number, we must multiply by 3/2.)
The figure below shows the time evolution of the interface on a slope of $\pi/4$ for Reynolds numbers of zero (red line) and $4\sin\alpha$ (green line). The perturbation wavenumber is $K = 0.1$ and the interface rapidly develops waves that grow as they are convected downslope for the higher Reynolds number, but decay when $Re = 0$.

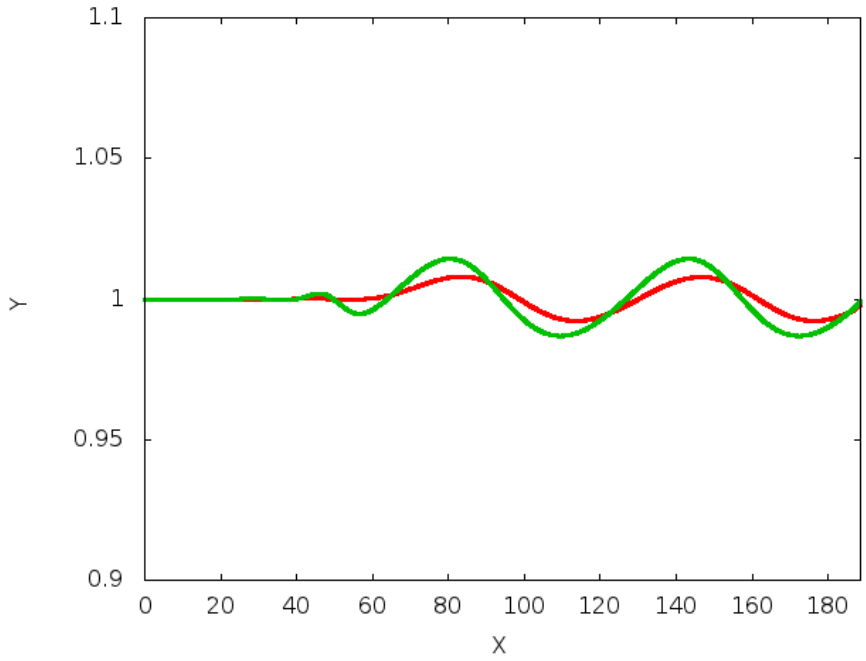


Figure 1.2 Time evolution (or static snapshot at t = 7.5) of the interface shape for Re = 0 (Red) and 4 sin(alpha) (Green).

The decay rate of the interfacial perturbation at $Re = 0$ is slow, but can be seen in the next figure, which shows the height of the interface at the downstream end of the domain plotted against time. The domain is chosen so that it will contain three waves and the decay or growth of successive crests and troughs can be seen.

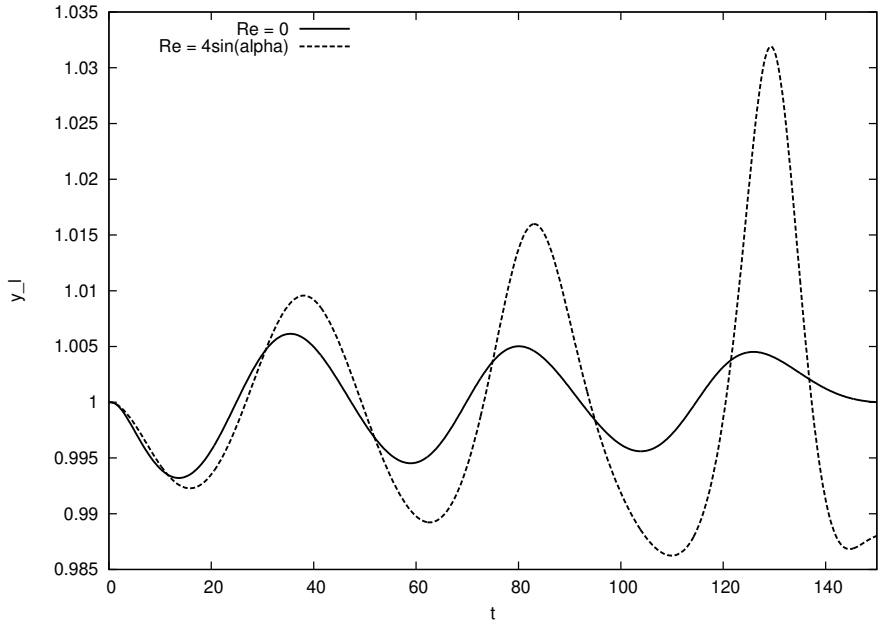


Figure 1.3 Time history of the interface position at the downstream end of the computational domain.

1.1.1 A note on the boundary conditions

Resolving the above analytic solution in a finite computational domain requires some thought about boundary conditions. We are only ever free to set one pressure value and setting the external pressure to zero fixes the pressure within the fluid. The boundary conditions at the plane are those of no-slip and at the free-surface the usual dynamic and kinematic conditions apply. Nonetheless, we have a number of possibilities for the boundary conditions at the "artificial" upstream and downstream computational boundaries.

- Prescribe periodic boundary conditions.
- Prescribe the velocity profile as a Dirichlet condition at both ends.
- Prescribe the appropriate hydrostatic pressure gradient and zero normal velocity.

We have chosen the last option, in which case the hydrostatic pressure gradient must be consistent with the external pressure. In other words, the pressure must be zero at the free surface ($y = 0$). Changing the external pressure would correspond to changing the film thickness, so the external pressure is directly responsible for enforcing a specific volume constraint, unless $\alpha = \pi/2$. When $\alpha = \pi/2$ there is no variation in hydrostatic pressure through the film and its thickness is not specified by the external pressure.
We must also worry about the boundary conditions on the free surface itself and we choose to impose a contact angle condition of $\pi/2$ at the upstream end, which ensures that the film remains flat. At the downstream end, we add a line tension term that arises from use of the surface divergence theorem to integrate the contribution of the dynamic boundary condition. This term can be used to enforce contact angle conditions in a weak formulation, but here we simply add the term using the angle calculated from the current position of the free surface.

1.2 Global parameters and functions

The global parameters are the Reynolds number, the dimensionless grouping Re/Fr, the angle of inclination of the slope α, the direction of the gravity vector G and the capillary number Ca, which only influences the dynamics.

```
namespace Global_Physical_Variables
{
 /// Reynolds number, based on the average velocity within the fluid film
 double Re=0.0;

 /// The product of Reynolds number and inverse Froude number
 /// is set to two in this problem, which gives the free surface velocity
 /// to be sin(alpha). [Set to three in order to get the same scale as
 /// used by Yih, Benjamin, etc]
 double ReInvFr=2.0;

 /// Angle of incline of the slope (45 degrees)
 double Alpha = 1.0*atan(1.0);

 /// The Vector direction of gravity, set in main()
 Vector<double> G(2,0.0);

 /// The Capillary number
 double Ca= 1.0;
```

The hydrostatic pressure field is specified as an applied traction. At the outlet (inlet), the outer unit normal is in the positive (negative) x direction and so the required traction is given by $-p$ (p),

$$\boldsymbol{t}|_{\text{outlet}} = (-(Re/Fr)\cos\alpha(1-y), 0), \qquad \boldsymbol{t}|_{\text{inlet}} = ((Re/Fr)\cos\alpha(1-y), 0).$$

These tractions are specified by the two different functions

```
 /// Function that prescribes the hydrostatic pressure field at the outlet
 void hydrostatic_pressure_outlet(const double& time, const Vector<double> &x,
                                  const Vector<double> &n,
                                  Vector<double> &traction)
 {
  traction[0] = ReInvFr*G[1]*(1.0 - x[1]);
  traction[1] = 0.0;
 }

 /// Function that prescribes hydrostatic pressure field at the inlet
 void hydrostatic_pressure_inlet(const double& time, const Vector<double> &x,
                                 const Vector<double> &n,
                                 Vector<double> &traction)
 {
  traction[0] = -ReInvFr*G[1]*(1.0 - x[1]);
  traction[1] = 0.0;
 }
 //end of traction functions
```

Note that `G[1]` is the component of the gravitational body force in the vertical direction, so $G[1] = -\cos\alpha$.

We must also specify the direction of the normals (directed out of the fluid) to the notional walls that form the inlet and outlet and a contact angle of $\pi/2$ that will be used as a boundary condition on the free surface at the upstream end of the domain. In this case the normal to the inlet is in the negative x-direction and the normal to the outlet is in the positive x-direction. The actual value of the `Wall_normal` vector is set in `main()`

```
 /// Direction of the wall normal vector (at the inlet)
 Vector<double> Wall_normal;
```

```
/// Function that specifies the wall unit normal at the inlet
void wall_unit_normal_inlet_fct(const Vector<double> &x,
                                Vector<double> &normal)
{
 normal=Wall_normal;
}

/// Function that specified the wall unit normal at the outlet
void wall_unit_normal_outlet_fct(const Vector<double> &x,
                                 Vector<double> &normal)
{
 //Set the normal
 normal = Wall_normal;
 //and flip the sign
 unsigned n_dim = normal.size();
 for(unsigned i=0;i<n_dim;++i) {normal[i] *= -1.0;}
}

/// The contact angle that is imposed at the inlet (pi)
double Inlet_Angle = 2.0*atan(1.0);
```

1.3 The driver code

We start by specifying the constitutive law used to define the mesh motion when pseudo-elastic deformation is used.

```
//start of main
int main(int argc, char **argv)
{
 using namespace Global_Physical_Variables;
 //Set the constitutive law for the mesh deformation
 Constitutive_law_pt  = new GeneralisedHookean(&Global_Physical_Variables::Nu);
```

Next, the type of fluid element is chosen according to specified compiler flags

```
#ifdef CR_ELEMENT
#define FLUID_ELEMENT QCrouzeixRaviartElement<2>
#else
#define FLUID_ELEMENT QTaylorHoodElement<2>
#endif
```

We then initialise the physical parameters, the Reynolds number and the direction of the gravitational body force, both based on the angle of inclination α.

```
 //Initialise physical parameters
 //Scale Reynolds number to be independent of alpha.
 Re = 4.0/sin(Alpha);

 //Set the direction of gravity
 G[0] = sin(Alpha);
 G[1] = -cos(Alpha);
```

We also set the direction of the notional wall normal vector.

```
 //The wall normal to the inlet is in the negative x direction
 Wall_normal.resize(2);
 Wall_normal[0] = -1.0;
 Wall_normal[1] = 0.0;
```

We now create the spine version of the problem, solve the steady problem, assign initial conditions by assuming that the problem has been at the steady state for all previous times, and then evolve the system in time.

```
 //Spine problem
 {
  //Create the problem
  SpineInclinedPlaneProblem<SpineElement<FLUID_ELEMENT >, BDF<2> >
   problem(30,4,Global_Physical_Variables::Length);

  //Solve the steady problem
  problem.solve_steady();

  //Prepare the problem for timestepping
  //(assume that it's been at the flat-film solution for all previous time)
  double dt = 0.1;
  problem.assign_initial_values_impulsive(dt);

  //Timestep it
  problem.timestep(dt,2);
 } //End of spine problem
```

Finally, exactly the same procedure is performed for the elastic problem

```
 //Elastic problem
 {
  //Create the problem
  ElasticInclinedPlaneProblem<
   PseudoSolidNodeUpdateElement<FLUID_ELEMENT,QPVDElement<2,3> >, BDF<2> >
     problem(30,4,Global_Physical_Variables::Length);

  //Solve the steady problem
  problem.solve_steady();
```

```
  //Prepare the problem for timestepping
  //(assume that it's been at the flat-film solution for all previous time)
  double dt = 0.1;
  problem.assign_initial_values_impulsive(dt);

  //Timestep it
  problem.timestep(dt,2);
 } //End of elastic problem
```

1.4 The mesh classes

The base mesh class is the `SimpleRectangularQuadMesh`: boundary 0 will be the wall; boundary 2 will be the free surface; and the remaining boundaries will be the inlet (3) and outlet (1). Below we shall demonstrate how to convert an existing mesh into a `SpineMesh` and `ElasticMesh` suitable for free-surface problems.

1.4.1 Creating the spine mesh

The `SpineInclinedPlaneMesh` inherits from the generic `SimpleRectangularQuadMesh` and adds vertical spines to the Nodes within the mesh in the constructor. Note that the resulting mesh is essentially the same as the `SingleLayerSpineMesh`, but has a somewhat simpler interface.

```
/// Create a spine mesh for the problem
//======================================================================
template <class ELEMENT>
class SpineInclinedPlaneMesh :
 public SimpleRectangularQuadMesh<ELEMENT>,
 public SpineMesh
{
public:
 SpineInclinedPlaneMesh(const unsigned &nx, const unsigned &ny,
                        const double &lx, const double &ly,
                        TimeStepper* time_stepper_pt) :
  SimpleRectangularQuadMesh<ELEMENT>
 (nx,ny,lx,ly,time_stepper_pt), SpineMesh()
  {
   //Find the number of linear points in the element
   unsigned n_p =  dynamic_cast<ELEMENT*>(finite_element_pt(0))->nnode_1d();
   //Reserve storage for the number of spines
   Spine_pt.reserve((n_p-1)*nx + 1);

   //Create single pointer to a spine
   Spine* new_spine_pt=0;

   //Now loop over the elements horizontally
   for(unsigned long j=0;j<nx;j++)
    {
     //In most elements, we don't assign a spine to the last column,
     //beacuse that will be done by the next element
     unsigned n_pmax = n_p-1;
     //In the last element, however, we must assign the final spine
     if(j==nx-1) {n_pmax = n_p;}

     //Loop over all nodes horizontally
     for(unsigned l2=0;l2<n_pmax;l2++)
      {
       //Create a new spine with unit height and add to the mesh
       new_spine_pt=new Spine(1.0);
       Spine_pt.push_back(new_spine_pt);

       // Get the node
       SpineNode* nod_pt=element_node_pt(j,l2);
       //Set the pointer to spine
       nod_pt->spine_pt() = new_spine_pt;
       //Set the fraction
       nod_pt->fraction() = 0.0;
       // Pointer to the mesh that implements the update fct
       nod_pt->spine_mesh_pt() = this;

       //Loop vertically along the spine
       //Loop over the elements
       for(unsigned long i=0;i<ny;i++)
        {
         //Loop over the vertical nodes, apart from the first
         for(unsigned l1=1;l1<n_p;l1++)
          {
           // Get the node
           SpineNode* nod_pt=element_node_pt(i*nx+j,l1*n_p+l2);
           //Set the pointer to the spine
           nod_pt->spine_pt() = new_spine_pt;
           //Set the fraction
           nod_pt->fraction()=(double(i)+double(l1)/double(n_p-1))/double(ny);
           // Pointer to the mesh that implements the update fct
```

```
           nod_pt->spine_mesh_pt() = this;
          }
         }
        }
      } //End of horizontal loop over elements
  } //end of constructor
```

In addition, a `spine_node_update()` function must be provided that determines how the `Nodes` move as functions of the `Spines`.

```
/// General node update function implements pure virtual function
/// defined in SpineMesh base class and performs specific node update
/// actions:  along vertical spines
virtual void spine_node_update(SpineNode* spine_node_pt)
 {
  //Get fraction along the spine
  double W = spine_node_pt->fraction();
  //Get spine height
  double H = spine_node_pt->h();
  //Set the value of y
  spine_node_pt->x(1) = W*H;
 }
```

1.4.2 Creating the ElasticMesh

The `ElasticInclinedPlaneMesh` inherits from the `SimpleRectangularQuadMesh` and the undeformed (reference) configuration is set to be the current position of the `Nodes`.

```
/// Create an Elastic mesh for the problem
//======================================================================
template <class ELEMENT>
class ElasticInclinedPlaneMesh :
 public SimpleRectangularQuadMesh<ELEMENT>,
 public SolidMesh
{
 //Public functions
 public:
 ElasticInclinedPlaneMesh(const unsigned &nx, const unsigned &ny,
                          const double &lx, const double &ly,
                          TimeStepper* time_stepper_pt) :
  SimpleRectangularQuadMesh<ELEMENT>(nx,ny,lx,ly,time_stepper_pt), SolidMesh()
  {
   //Make the current configuration the undeformed one
   set_lagrangian_nodal_coordinates();
  }
};
```

Note that the specification of the ElasticMesh is much simpler than that of a SpineMesh because no decision needs to be taken about how to describe the motion using Spines.

1.5 The problem classes

1.5.1 The generic problem

For ease of exposition, all generic functionality is included in the `InclinedPlaneProblem` class, which is templated by the bulk `ELEMENT` and the `INTERFACE_ELEMENT`. The class includes storage for the different sub-meshes: Bulk, the Traction elements associated with the inlet and outlet, the (free) Surface elements and the point elements associated with the ends of the interface. In addition, a string `Output_prefix` is used to distinguish between the output files from different formulations.

```
/// Generic problem class that will form the base class for both
/// spine and elastic mesh-updates of the problem.
/// Templated by the bulk element and interface element types
//======================================================================
template<class ELEMENT, class INTERFACE_ELEMENT>
class InclinedPlaneProblem : public Problem
{

protected:

 /// Bulk fluid mesh
 Mesh* Bulk_mesh_pt;

 /// Mesh for the traction elements that are added at inlet and outlet
 Mesh* Traction_mesh_pt;

 /// Mesh for the free surface elements
 Mesh* Surface_mesh_pt;

 /// Mesh for the point elements at each end of the free surface
 Mesh* Point_mesh_pt;

 /// Prefix for output files
```

```
 std::string Output_prefix;
```

The time-dependent perturbation is introduced in the function `actions_before_implicit_timestep()`, which sets the vertical velocity on the wall (boundary 0)

$$v = \epsilon \sin(Kx) t e^{-t}$$

```
void actions_before_implicit_timestep()
 {
  //Read out the current time
  double time = this->time_pt()->time();
  //Now add a temporary sinusoidal suction and blowing to the base
  //Amplitude of the perturbation
  double epsilon = 0.01;
  //Loop over the nodes on the base
  unsigned n_node = this->Bulk_mesh_pt->nboundary_node(0);
  for(unsigned n=0;n<n_node;n++)
   {
    Node* nod_pt = this->Bulk_mesh_pt->boundary_node_pt(0,n);
    double arg = Global_Physical_Variables::K*nod_pt->x(0);
    double value = sin(arg)*epsilon*time*exp(-time);
    nod_pt->set_value(1,value);
   }
 } //end_of_actions_before_implicit_timestep
```

The function `make_traction_elements()` creates `NavierStokesTractionElement` s adjacent to the mesh boundaries 3 (the inlet) and 1 (the inlet). These elements are added to the `Mesh Traction_mesh_↩ pt`, which is itself constructed in the function and pointers to the appropriate traction functions are assigned.

```
/// Function to add the traction boundary elements to boundaries
/// 3(inlet) and 1(outlet) of the mesh
void make_traction_elements()
 {
  //Create a new (empty mesh)
  Traction_mesh_pt = new Mesh;
  //Inlet boundary conditions (boundary 3)
  {
   unsigned b = 3;
   //Find the number of elements adjacent to mesh boundary
   unsigned n_boundary_element = Bulk_mesh_pt->nboundary_element(b);
   //Loop over these elements and create the traction elements
   for(unsigned e=0;e<n_boundary_element;e++)
    {
     NavierStokesTractionElement<ELEMENT> *surface_element_pt =
      new NavierStokesTractionElement<ELEMENT>
      (Bulk_mesh_pt->boundary_element_pt(b,e),
       Bulk_mesh_pt->face_index_at_boundary(b,e));
     //Add the elements to the mesh
     Traction_mesh_pt->add_element_pt(surface_element_pt);
     //Set the traction function
     surface_element_pt->traction_fct_pt() =
      &Global_Physical_Variables::hydrostatic_pressure_inlet;
    }
  }

  //Outlet boundary conditions (boundary 1)
  {
   unsigned b=1;
   //Find the number of elements adjacent to mesh boundary
   unsigned n_boundary_element = Bulk_mesh_pt->nboundary_element(b);
   //Loop over these elements and create the traction elements
   for(unsigned e=0;e<n_boundary_element;e++)
    {
     NavierStokesTractionElement<ELEMENT> *surface_element_pt =
      new NavierStokesTractionElement<ELEMENT>
      (Bulk_mesh_pt->boundary_element_pt(b,e),
       Bulk_mesh_pt->face_index_at_boundary(b,e));
     //Add the elements to the mesh
     Traction_mesh_pt->add_element_pt(surface_element_pt);
     //Set the traction function
     surface_element_pt->traction_fct_pt() =
      &Global_Physical_Variables::hydrostatic_pressure_outlet;
    }
  }
 } //end of make_traction_elements
```

The function `make_free_surface_elements()` creates the appropriate `INTERFACE_ELEMENTs` adjacent to the free surface (boundary 2), sets the capillary number and also creates free-surface boundary elements at the left- and right-hand ends of the interface. If these "point" elements are not included then the surface tension is not applied correctly at the edges of the domain. The contact angle is set to be the value `Inlet_Angle` at the left-hand edge of the domain.

```
//Make the free surface elements on the top surface
void make_free_surface_elements()
 {
  //Create the (empty) meshes
  Surface_mesh_pt = new Mesh;
```

```
  Point_mesh_pt = new Mesh;

  //The free surface is on the boundary 2
  unsigned b = 2;
  unsigned n_boundary_element = Bulk_mesh_pt->nboundary_element(b);
  //Loop over the elements and create the appropriate interface elements
  for(unsigned e=0;e<n_boundary_element;e++)
   {
    INTERFACE_ELEMENT *surface_element_pt =
     new INTERFACE_ELEMENT
     (Bulk_mesh_pt->boundary_element_pt(b,e),
      Bulk_mesh_pt->face_index_at_boundary(b,e));
    //Add elements to the mesh
    Surface_mesh_pt->add_element_pt(surface_element_pt);
    //Assign the capillary number to the free surface
    surface_element_pt->ca_pt() =
     &Global_Physical_Variables::Ca;

    //Make a point element from left-hand side of the
    //first surface element (note that this relies on knowledge of
    //the element order within the mesh)
    if(e==0)
     {
      FluidInterfaceBoundingElement* point_element_pt =
       surface_element_pt->make_bounding_element(-1);
      //Add element to the point mesh
      Point_mesh_pt->add_element_pt(point_element_pt);
      //Set the capillary number
      point_element_pt->ca_pt() = &Global_Physical_Variables::Ca;
      //Set the wall normal
      point_element_pt->wall_unit_normal_fct_pt() =
       &Global_Physical_Variables::wall_unit_normal_inlet_fct;
      //Set the contact angle (using the strong version of the constraint)
      point_element_pt->set_contact_angle(
       &Global_Physical_Variables::Inlet_Angle);
     }

    //Make another point element from the right-hand side of the
    //last surface element (note that this relies on knowledge of
    //the element order within the mesh)
    if(e==n_boundary_element-1)
     {
      FluidInterfaceBoundingElement* point_element_pt =
       surface_element_pt->make_bounding_element(1);
      //Add element to the mesh
      Point_mesh_pt->add_element_pt(point_element_pt);
      //Set the capillary number
      point_element_pt->ca_pt() = &Global_Physical_Variables::Ca;
      // Set the function that specifies the wall normal
      point_element_pt->wall_unit_normal_fct_pt() =
       &Global_Physical_Variables::wall_unit_normal_outlet_fct;
     }
   }
  } //end of make_free_surface_elements
```

The function `complete_build()` assigns physical parameters to the fluid elements, sets the boundary conditions and assigns equation numbers.

```
 void complete_build()
  {
   using namespace Global_Physical_Variables;

   //Complete the build of the fluid elements by passing physical parameters
   //Find the number of bulk elements
   unsigned n_element = Bulk_mesh_pt->nelement();
   //Loop over all the fluid elements
   for(unsigned e=0;e<n_element;e++)
    {
     //Cast to a fluid element
     ELEMENT *temp_pt = dynamic_cast<ELEMENT*>(Bulk_mesh_pt->element_pt(e));

     //Set the Reynolds number
     temp_pt->re_pt() = &Re;
     //The Strouhal number is 1, so ReSt = Re
     temp_pt->re_st_pt() = &Re;
     //Set the Reynolds number / Froude number
     temp_pt->re_invfr_pt() = &ReInvFr;
     //Set the direction of gravity
     temp_pt->g_pt() = &G;
    }

   //------------Set the boundary conditions for this problem----------

   {
    //Determine whether we are solving an elastic problem or not
    bool elastic = false;
```

```
    if(dynamic_cast<SolidNode*>(Bulk_mesh_pt->node_pt(0))) {elastic=true;}

    //Loop over the bottom of the mesh (the wall of the channel)
    unsigned n_node = Bulk_mesh_pt->nboundary_node(0);
    for(unsigned j=0;j<n_node;j++)
     {
      //Pin the u- and v- velocities
      Bulk_mesh_pt->boundary_node_pt(0,j)->pin(0);
      Bulk_mesh_pt->boundary_node_pt(0,j)->pin(1);

      //If we are formulating the elastic problem pin both positions
      //of nodes
      if(elastic)
       {
        static_cast<SolidNode*>(Bulk_mesh_pt->boundary_node_pt(0,j))
         ->pin_position(0);
        static_cast<SolidNode*>(Bulk_mesh_pt->boundary_node_pt(0,j))
         ->pin_position(1);
       }
     }

    //Loop over the inlet and set the Dirichlet condition
    //of no vertical velocity
    n_node = Bulk_mesh_pt->nboundary_node(3);
    for(unsigned j=0;j<n_node;j++)
     {
      Bulk_mesh_pt->boundary_node_pt(3,j)->pin(1);

      //If elastic pin horizontal position of nodes
      if(elastic)
       {
        static_cast<SolidNode*>(Bulk_mesh_pt->boundary_node_pt(3,j))
         ->pin_position(0);
       }
     }

    //Loop over the outlet and set the Dirichlet condition
    //of no vertical velocity
    n_node = Bulk_mesh_pt->nboundary_node(1);
    for(unsigned j=0;j<n_node;j++)
     {
      Bulk_mesh_pt->boundary_node_pt(1,j)->pin(1);

      //If elastic pin horizontal position
      if(elastic)
       {
        static_cast<SolidNode*>(Bulk_mesh_pt->boundary_node_pt(1,j))
         ->pin_position(0);
       }
     }
   }
   //Attach the boundary conditions to the mesh
   std::cout << assign_eqn_numbers() << " in the main problem" << std::endl;
  } //end of complete_build
```

Note that boundary conditions for the nodal positions in the pseudo-elastic formulation are specified by testing whether the `Nodes` are `SolidNodes`. In this case, the `Nodes` on the inlet and outlet boundaries are constrained to remain at the same horizontal position and the `Nodes` on the plane wall are fixed.

The function `solve_steady()` initialises the velocity of at all `Nodes` to the flat-film solution, solves the steady equations and writes the solution to a file.

```
void InclinedPlaneProblem<ELEMENT,INTERFACE_ELEMENT>::solve_steady()
{
 //Load the namespace
 using namespace Global_Physical_Variables;
 //Initially set all nodes to the Nusselt flat-film solution
 {
  unsigned n_node = Bulk_mesh_pt->nnode();
  for(unsigned  n=0;n<n_node;n++)
   {
    double y = Bulk_mesh_pt->node_pt(n)->x(1);
    //Top row
    Bulk_mesh_pt->node_pt(n)->set_value(0,0.5*ReInvFr*sin(Alpha)*(2.0*y - y*y));
   }
 }
 //Do one steady solve
 steady_newton_solve();

 //Output the full flow field
 std::string filename = Output_prefix;;
 filename.append("_output.dat");
 ofstream file(filename.c_str());
 Bulk_mesh_pt->output(file,5);
 file.close();
} //end of solve_steady
```

Finally, the function `timestep()` takes a number of fixed timesteps writing vertical positions and the time to a

trace file and writing the complete flow field to disk after a given number of timesteps.

```
timestep(const double &dt, const unsigned &n_tsteps)
{
 //Need to use the Global variables here
 using namespace Global_Physical_Variables;
 //Open an output file
 std::string filename = Output_prefix;
 filename.append("_time_trace.dat");
 ofstream trace(filename.c_str());
 //Counter that will be used to output the full flowfield
 //at certain timesteps
 int counter=0;
 //Initial output of the time and the value of the vertical position at the
 //left and right-hand end of the free surface
 trace « time_pt()->time() « " "
       « Bulk_mesh_pt->boundary_node_pt(2,0)->value(1)
       « " "
       «  Bulk_mesh_pt->
  boundary_node_pt(2, Bulk_mesh_pt->nboundary_node(2)-1)->x(1)
       « " "
       « std::endl;
 //Loop over the desired number of timesteps
 for(unsigned t=1;t<=n_tsteps;t++)
  {
   //Increase the counter
   counter++;
   cout « std::endl;
   cout « "--------------TIMESTEP " « t« " ------------------" « std::endl;

   //Take a timestep of size dt
   unsteady_newton_solve(dt);

   //Uncomment to get full solution output
   if(counter==2) //Change this number to get output every n steps
    {
     std::ofstream file;
     std::ostringstream filename;
     filename « Output_prefix « "_step" « Re « "_" « t « ".dat";
     file.open(filename.str().c_str());
     Bulk_mesh_pt->output(file,5);
     file.close();

     counter=0;
    }

   //Always output the interface
   {
    std::ofstream file;
     std::ostringstream filename;
     filename « Output_prefix « "_interface_" « Re « "_" « t « ".dat";
     file.open(filename.str().c_str());
     Surface_mesh_pt->output(file,5);
     file.close();
   }

   //Output the time and value of the vertical position of the free surface
   //at the left- and right-hand ends
   trace « time_pt()->time() « " "
         « Bulk_mesh_pt->boundary_node_pt(2,0)->x(1) « " "
         «
    Bulk_mesh_pt->
    boundary_node_pt(2,Bulk_mesh_pt->nboundary_node(2)-1)->x(1) « " "
         « std::endl;
  }
} //end of timestep
```

1.5.2 The spine-based formulation

The class `SpineInclinedPlaneProblem` inherits from the generic `InclinedPlaneProblem` class and requires only minor modification. The constructor sets the string `Output_prefix`, builds a timestepper, builds the specific `SpineMesh`, creates the appropriate `FaceElements`, adds all sub-meshes to the `Problem`, builds the global mesh and then calls `InclinedPlaneProblem::complete_build()`.

```
 SpineInclinedPlaneProblem(const unsigned &nx, const unsigned &ny,
                           const double &length):
  InclinedPlaneProblem<ELEMENT,SpineLineFluidInterfaceElement<ELEMENT> >
  (nx,ny,length)
  {
   //Set the name
   this->Output_prefix = "spine";

   //Create our one and only timestepper, with adaptive timestepping
   this->add_time_stepper_pt(new TIMESTEPPER);

   //Create the bulk mesh
```

```
  this->Bulk_mesh_pt = new  SpineInclinedPlaneMesh<ELEMENT>(
   nx,ny,length,1.0,this->time_stepper_pt());

  //Create the traction elements
  this->make_traction_elements();
  //Create the free surface elements
  this->make_free_surface_elements();

  //Add all sub meshes to the problem
  this->add_sub_mesh(this->Bulk_mesh_pt);
  this->add_sub_mesh(this->Traction_mesh_pt);
  this->add_sub_mesh(this->Surface_mesh_pt);
  this->add_sub_mesh(this->Point_mesh_pt);
  //Create the global mesh
  this->build_global_mesh();

  //Complete the build of the problem
  this->complete_build();
 }
```

In a spine-based formulation, the nodal positions must be updated after every Newton step, which is achieved by overloading the function `Problem::actions_before_newton_convergence_check()`

```
/// Spine heights/lengths are unknowns in the problem so their
/// values get corrected during each Newton step. However,
/// changing their value does not automatically change the
/// nodal positions, so we need to update all of them
void actions_before_newton_convergence_check()
 {this->Bulk_mesh_pt->node_update();}
```

We also specify a destructor to clean up memory allocated by the class.

1.5.3 The pseudo-solid-based formulation

The class `ElasticInclinedPlaneProblem` inherits from the generic `InclinedPlaneProblem` class and also requires only minor modification. The constructor sets the string `Output_prefix`, builds a timestepper, builds the specific `SolidMesh`, sets the constitutive law for the bulk elements, creates the appropriate `FaceElements`, adds all sub-meshes to the `Problem`, builds the global mesh and then calls `InclinedPlaneProblem::complete_build()`

```
ElasticInclinedPlaneProblem(const unsigned &nx, const unsigned &ny,
                            const double &length) :
 InclinedPlaneProblem<ELEMENT,ElasticLineFluidInterfaceElement<ELEMENT> >
 (nx,ny,length)
 {
  //Set the name
  this->Output_prefix = "elastic";

  //Create our one and only timestepper, with adaptive timestepping
  this->add_time_stepper_pt(new TIMESTEPPER);

  //Create the bulk mesh
  this->Bulk_mesh_pt = new  ElasticInclinedPlaneMesh<ELEMENT>(
   nx,ny,length,1.0,this->time_stepper_pt());

  //Set the consititutive law for the elements
  unsigned n_element = this->Bulk_mesh_pt->nelement();
  //Loop over all the fluid elements
  for(unsigned e=0;e<n_element;e++)
   {
    //Cast to a fluid element
    ELEMENT *temp_pt = dynamic_cast<ELEMENT*>(
     this->Bulk_mesh_pt->element_pt(e));
    //Set the constitutive law
    temp_pt->constitutive_law_pt() =
     Global_Physical_Variables::Constitutive_law_pt;
   }

  //Create the traction elements
  this->make_traction_elements();
  //Create the free surface element
  this->make_free_surface_elements();

  //Add all sub meshes to the problem
  this->add_sub_mesh(this->Bulk_mesh_pt);
  this->add_sub_mesh(this->Traction_mesh_pt);
  this->add_sub_mesh(this->Surface_mesh_pt);
  this->add_sub_mesh(this->Point_mesh_pt);
  //Create the global mesh
  this->build_global_mesh();

  //Complete the rest of the build
  this->complete_build();
 } //end of constructor
```

In a pseudo-solid formulation, it is advantageous to reset the undeformed configuration after every timestep (an up-

dated Lagrangian formulation). Hence, the `Problem::actions_after_implicit_timestep()` function is overloaded

```
void actions_after_implicit_timestep()
 {
  //Now loop over all the nodes and reset their Lagrangian coordinates
  unsigned n_node = this->Bulk_mesh_pt->nnode();
  for(unsigned n=0;n<n_node;n++)
   {
    //Cast node to an elastic node
    SolidNode* temp_pt =
     static_cast<SolidNode*>(this->Bulk_mesh_pt->node_pt(n));
    for(unsigned j=0;j<2;j++) {temp_pt->xi(j) = temp_pt->x(j);}
   }
 } //end of actions_after_implicit_timestep
```

We also specify a destructor to clean up memory allocated by the class.

1.6 Exercises

1. Confirm that the steady solution agrees with the exact solution.
2. Investigate what happens when the angle is varied. What happens when the angle is set to zero? What happens when the angle is set to $\pi/2$?
3. What happens if the hydrostatic pressure boundary conditions are not applied?
4. How does the stability of the system to the perturbation change with angle, Ca and K? Are the results in agreement with the theoretical predictions?
5. Are the results independent of the length of the domain?
6. Compare the spine-based and pseudo-elastic-based formulations? What is the same and what is different? Which method do you prefer?

1.7 Source files for this tutorial

- The source files for this tutorial are located in the directory:

  demo_drivers/navier_stokes/inclined_plane/

- The driver code is:

  demo_drivers/navier_stokes/inclined_plane/inclined_plane.cc

1.8 PDF file

A pdf version of this document is available. \